

Window bounds for repetitions and block complexity of algebraic numbers

Abstract

Let ξ be a real algebraic irrational number and $b \geq 2$ an integer. Write $R(L)$ for the length of the shortest prefix of its fractional base- b expansion containing two occurrences of some word of length L , and $p(L)$ for its block complexity. For every fixed $\theta > 1$ there is $c > 0$ such that every sufficiently large interval $[X, X^\theta]$ contains an integer L for which

$$R(L) > cL \sqrt{\frac{\log L}{\log \log L}}, \quad p(L) > \frac{c}{2}L \sqrt{\frac{\log L}{\log \log L}}.$$

Consequently the logarithmically weighted complexity limsup is infinite for every weight exponent $\gamma > 1/2$. We also obtain windows $[X, X^{1+o(1)}]$ below this endpoint. The arithmetic input is a finite-support moving-weight sparse-parameter theorem proved here by adapting the auxiliary-polynomial argument of Evertse and Ferretti. The additional combinatorial step retains all dyadic prefix scales: when several scales yield the same approximation scale, their multiplicity gives an exponential improvement in approximation quality. Summing fixed-quality counting bounds removes a logarithmic loss without requiring an extension with a different smallness exponent at each point.

1 Introduction

Fix a real algebraic irrational number ξ and an integer $b \geq 2$. We may assume $0 < \xi < 1$, and write $\xi = \sum_{i \geq 1} d_i b^{-i}$, with $d_i \in \{0, \dots, b-1\}$. Define

$$p(L) = \#\{d_{j+1} \cdots d_{j+L} : j \geq 0\}.$$

Following Bugeaud and Kim [4], let $R(L)$ be the least prefix length containing two, possibly overlapping, occurrences of a word of length L . We use R rather than r to distinguish this function from the preperiod length of a rational approximation. Pigeonholing gives

$$R(L) \leq L + p(L). \tag{1.1}$$

Indeed, the prefix of length $L + p(L)$ contains $p(L) + 1$ starting positions for length- L words. All logarithms without a subscript are natural, and iterated logarithms are used only for sufficiently large arguments.

Theorem 1.1. *For every fixed $\theta > 1$, there is a constant $c = c(\xi, b, \theta) > 0$ such that, for every sufficiently large real X , there is an integer $L \in [X, X^\theta]$ satisfying*

$$R(L) > cL \sqrt{\frac{\log L}{\log \log L}}, \quad p(L) > \frac{c}{2}L \sqrt{\frac{\log L}{\log \log L}}. \tag{1.2}$$

Corollary 1.2. *For each $g \in \{R, p\}$,*

$$\limsup_{L \rightarrow \infty} \frac{g(L) \sqrt{\log \log L}}{L \sqrt{\log L}} > 0.$$

For every $\gamma > 1/2$ and every $\eta < 1/2$,

$$\limsup_{L \rightarrow \infty} \frac{g(L) (\log \log L)^\gamma}{L \sqrt{\log L}} = \infty, \quad \limsup_{L \rightarrow \infty} \frac{g(L)}{L (\log L)^\eta} = \infty. \quad (1.3)$$

The repetition-function statement is proved directly. Inequality (1.1) then transfers its lower bound to p ; a lower bound for p alone would not imply the repetition-function statement.

Theorem 1.3. *Let $F(x) = (\log x)^u (\log \log x)^{-\gamma}$, where $u > 0$, $\gamma \geq 0$, and either $u < 1/2$, or $u = 1/2$ and $\gamma > 1/2$. For every $C > 0$ there is $A = A(\xi, b, C, u, \gamma) > 0$ such that, for every sufficiently large X , the interval*

$$\left[X, X \exp\{AF(X)^2 \log(2F(X))\} \right] \quad (1.4)$$

contains an integer L with both $R(L) > CLF(L)$ and $p(L) > CLF(L)$. Its upper endpoint is $X^{1+o(1)}$.

Adamczewski and Bugeaud proved $p(L)/L \rightarrow \infty$ for algebraic irrationals in integer bases [1]. Bugeaud and Evertse proved the logarithmic limsup bound with $\eta < 1/11$ [2, Theorem 2.1]. The arithmetic argument here uses their repetition method and the semistable machinery of [3, Sections 8–14]. A fixed-weight theorem alone does not permit the exact ratio $\rho = r/t$ to vary from point to point. Section 5 proves the needed finite-support variant, including a bound for its height threshold; Section 4 checks semistability for every weight in the relevant family.

The subsequent counting argument uses that theorem separately at each fixed quality. If μ dyadic prefix scales yield a single dyadic approximation scale, the largest prefix yields quality at least a constant times $2^\mu/B$. Summation over the multiplicity levels gives a bound $O(B^2 \log(2B))$ for the original number of scales. This replaces greedy thinning of the prefix scales. The proof of this step is in Section 6.

Constants may depend on ξ, b , and on the fixed window or function parameters explicitly indicated. No effective numerical starting length or assertion uniform in varying algebraic numbers or bases is made. The endpoint statement is a positive limsup; it does not assert a positive lower bound at every sufficiently large length.

2 From repetitions to approximations

Lemma 2.1. *Suppose that the prefix of length H contains two occurrences of a word of length k . Then there are integers r, s, t, v, a such that*

$$t = r + s, \quad 0 \leq r < t, \quad t + v \leq H, \quad s \geq v \geq k/3, \quad 0 \leq a \leq b^t - b^r, \quad (2.1)$$

and

$$|(b^t - b^r)\xi - a| \leq b^{-v}. \quad (2.2)$$

Proof. Let D be the difference of the two starting positions. If $D \geq k$, the two occurrences are disjoint and we take $v = k$. If $D < k$, their union is a word of length $k + D$ with period D . Its first two blocks of length $v' = D \lfloor (k + D)/(2D) \rfloor$ are equal and disjoint. Here $v' \geq k/3$: if $D \geq k/3$, use $v' \geq D$; otherwise use $v' > (k - D)/2$. Thus in either case the prefix has a decomposition $UVWVX$, with $v = |V| \geq k/3$.

Put $r = |U|$, $s = |VW|$, $t = r + s$, and $A_j = \sum_{i=1}^j d_i b^{j-i}$, $A_0 = 0$. Set $a = A_t - A_r$. The number $a/(b^t - b^r)$ has expansion $U(VW)^\infty$ and agrees with ξ through the first $t + v$ digits. Its distance from ξ is at most b^{-t-v} , which proves (2.2). If P is the integer value of VW , then $a = (b^s - 1)A_r + P$, giving $0 \leq a \leq b^t - b^r$. The closed interval of values with a specified finite prefix also covers a periodic tail consisting entirely of the digit $b - 1$. The remaining length bounds follow from the disjoint occurrences. $\square$

Lemma 2.2 (Repetitions in a prescribed window). *Let $2 \leq X < Y$, let F be positive and nondecreasing on $[X, Y]$, and write $F_Y = F(Y)$. Suppose*

$$R(k) \leq CkF(k) \quad \text{for every integer } k \in [X, Y], \quad CF_Y \geq 1.$$

For each integer i with $8CF_Y X \leq 2^i \leq Y$, there is an approximation satisfying (2.1)–(2.2), with $H = 2^i$, such that

$$v_i \geq \frac{2^i}{24CF_Y}, \quad t_i \geq X/3. \quad (2.3)$$

Proof. Take $k_i = \lfloor 2^i/(4CF_Y) \rfloor$. The lower bound on 2^i gives $k_i \geq 2X - 1 \geq X$, and $CF_Y \geq 1$ gives $k_i \leq Y$. Thus $R(k_i) \leq CF_Y k_i \leq 2^i/4$, so the prefix of length 2^i contains the required repetition. Also $k_i \geq 2^i/(8CF_Y)$. Lemma 2.1 gives $v_i \geq k_i/3 \geq 2^i/(24CF_Y)$. Since $t_i \geq v_i$, the bound $t_i \geq X/3$ follows. All uses of the assumed upper bound for R occur at lengths inside $[X, Y]$. $\square$

3 Twisted heights and the exact weights

For a number field K write M_K for its places, normalized by $\|z\|_v = |z|_p^{d_v}$ on $z \in \mathbb{Q}$ when $v \mid p$, where $d_v = [K_v : \mathbb{Q}_p]/[K : \mathbb{Q}]$ and $\sum_{v \mid p} d_v = 1$. At an infinite place represented by an embedding σ_v the corresponding formula is $\|z\|_v = |\sigma_v(z)|^{d_v}$. For a system of linearly independent forms $\mathcal{L} = (L_{iv} : v \in M_K, 1 \leq i \leq 3)$ and real weights c_{iv} , put

$$H_{\mathcal{L}, c, Q}(x) = \prod_{v \in M_K} \max_{1 \leq i \leq 3} (\|L_{iv}(x)\|_v Q^{-c_{iv}}). \quad (3.1)$$

For $x \in \overline{\mathbb{Q}}^3$ use a finite extension E/K containing its coordinates, keep the forms unchanged at $w \mid v$, and multiply the weights by $d(w \mid v) = [E_w : K_v]/[E : K]$. This gives an extension-independent height.

Choose a finite Galois extension $K/\mathbb{Q}$ in $\mathbb{C}$ containing ξ , and let $G = \text{Gal}(K/\mathbb{Q})$ and $d = [\mathbb{Q}(\xi) : \mathbb{Q}] \geq 2$. Write $\beta_1, \dots, \beta_d$ for the distinct conjugates of ξ . They are all nonzero. For each infinite place v , choose $\sigma_v \in G$ with $\|z\|_v = |\sigma_v(z)|^{d_v}$, and set $\beta_v = \sigma_v^{-1}(\xi)$. Define the fixed forms by

$$(L_{1v}, L_{2v}, L_{3v}) = \begin{cases} (X_1, X_2, L_{\beta_v}), & v \mid \infty, \\ (X_1, X_2, X_3), & v \nmid \infty, \end{cases} \quad L_\beta = -\beta X_1 + \beta X_2 + X_3. \quad (3.2)$$

Their local determinants are all 1, and there are $d + 3$ distinct forms. For every rational vector x ,

$$\|L_{\beta_v}(x)\|_v = |\sigma_v(L_{\beta_v}(x))|^{d_v} = |L_\xi(x)|^{d_v}. \quad (3.3)$$

This is the inverse-conjugate transport used in [2, Section 5, equations (5.14)–(5.17)]. It uses smallness only at the specified real value of ξ .

Lemma 3.1 (Mass of a conjugate). *For every conjugate β of ξ ,*

$$\sum_{\substack{v|\infty \\ \beta_v=\beta}} d_v = \frac{1}{d}. \quad (3.4)$$

Proof. Let $\iota \in G$ be complex conjugation on K and $H = \langle \iota \rangle$. Infinite places are represented by the right cosets $H\sigma$. Since ξ is real, $(\iota\sigma)^{-1}(\xi) = \sigma^{-1}(\xi)$, so β_v does not depend on the representative. The field K is either totally real or totally imaginary, and $d_v = |H|/|G|$ in either case. The set of $\sigma \in G$ with $\sigma^{-1}(\xi) = \beta$ has cardinality $|G|/d$ by orbit–stabilizer. It is a union of right H -cosets, each of cardinality $|H|$. Its corresponding total place weight is therefore $(|G|/(d|H|))(|H|/|G|) = 1/d$. $\square$

For each rational prime $p \mid b$, put

$$\omega_p = \frac{\text{ord}_p(b) \log p}{\log b}, \quad \kappa_v = d_v \omega_p \quad (v \mid p). \quad (3.5)$$

These numbers are positive and satisfy $\sum_{p \mid b} \omega_p = 1$ and $\sum_{v \mid b} \kappa_v = 1$, where $v \mid b$ means $v \mid p$ for some $p \mid b$. Let S consist of all infinite places and all places above the prime divisors of b .

Fix $0 < \varepsilon \leq 1/2$ and $0 \leq \rho \leq 1 - \varepsilon$. In the order of (3.2), define raw weights by

$$e_v = \begin{cases} d_v(1, \rho, -\varepsilon), & v \mid \infty, \\ \kappa_v(-1, -\rho, 0), & v \mid b, \\ (0, 0, 0), & v \notin S. \end{cases} \quad (3.6)$$

Put

$$\bar{e}_v = \frac{1}{3} \sum_i e_{iv}, \quad D_\varepsilon = 1 + \varepsilon/3, \quad c_{iv}(\rho) = \frac{e_{iv} - \bar{e}_v}{D_\varepsilon}, \quad \delta = \frac{\varepsilon}{3 + \varepsilon}. \quad (3.7)$$

The two mass identities give

$$\sum_i c_{iv} = 0, \quad \sum_v \max_i c_{iv} = 1, \quad \sum_v \bar{e}_v = -\varepsilon/3. \quad (3.8)$$

Indeed $\sum_{v,i} e_{iv} = -\varepsilon$ and the sum of the raw local maxima is 1; centering changes the latter to $1 + \varepsilon/3$.

Consider any approximation satisfying (2.1)–(2.2). Choose $0 < \varepsilon \leq \min(v/t, 1/2)$ and put $\rho = r/t$. Since $t - r \geq v$, we have $\rho \leq 1 - \varepsilon$. Let $x = (b^t, b^r, a)$, $\Psi = b^t$, and $Q = \Psi^{D_\varepsilon}$. At the infinite places, (2.2) and (3.3) give $\|L_{iv}(x)\|_v \leq \Psi^{e_{iv}(\rho)}$. At a place $v \mid p \mid b$, the same bounds follow from

$$\|b^t\|_v = \Psi^{-\kappa_v}, \quad \|b^r\|_v = \Psi^{-\kappa_v \rho}, \quad \|a\|_v \leq 1.$$

At all other finite places the coordinates are integral. Hence

$$H_{\mathcal{L}, c(\rho), Q}(x) \leq \Psi^{\sum_v \bar{e}_v} = Q^{-\delta}, \quad \delta = \frac{\varepsilon}{3 + \varepsilon}. \quad (3.9)$$

For a collection of such approximations with a common ε , the condition $t_{j+1} > 2t_j$ implies $Q_{j+1} > Q_j^2$. No primitivity of x is needed: Ψ is an external parameter, not an identification with its projective height.

4 Uniform semistability

For a q -dimensional subspace $U \subset \overline{\mathbb{Q}}^3$, define its local weight as the minimum of $\sum_{i \in I} c_{iv}$ over all q -element sets I for which $(L_{iv}|_U)_{i \in I}$ is linearly independent. Its global weight $w_c(U)$ is the sum over places. The system is semistable if $w_c(U) \leq 0$ for every nonzero proper U . Define w_e analogously. Centering gives

$$w_c(U) = \frac{w_e(U) + q\varepsilon/3}{D_\varepsilon}. \quad (4.1)$$

Proposition 4.1. *For the family (3.6)–(3.7), all nonzero proper subspaces have negative weight. More precisely,*

$$w_c(U) \leq -\frac{\varepsilon}{6D_\varepsilon} \quad (\dim U = 1), \quad w_c(U) \leq -\frac{\varepsilon}{3D_\varepsilon} \quad (\dim U = 2).$$

Proof. By Lemma 3.1, the infinite-place contribution to $w_e(U)$ can be computed using each distinct L_β once, with raw weights $(1, \rho, -\varepsilon)/d$. The finite contributions combine into one minimum calculation with coordinate forms and weights $(-1, -\rho, 0)$, since $\sum_{v|b} \kappa_v = 1$. This grouping concerns only the subspace weight calculation; all original places and norms remain in the height.

Consider a line U . If no L_β vanishes on it, its infinite contribution is $-\varepsilon$ and its finite contribution is at most zero. If two distinct L_β vanish, subtracting them shows $U = \langle (1, 1, 0) \rangle$; all L_β then vanish and $w_e(U) = \rho - 1 \leq -\varepsilon$. Otherwise precisely one L_β vanishes. If $X_1|_U \neq 0$, the exceptional infinite contribution is at most $1/d$, the remaining infinite contribution is $-(1 - 1/d)\varepsilon$, and the finite contribution is -1 . Thus $w_e(U) \leq -(1 - 1/d)(1 + \varepsilon)$. If $X_1|_U = 0$, then $X_2|_U \neq 0$ and $w_e(U) = -(1 - 1/d)(\rho + \varepsilon)$. Since $d \geq 2$, every line therefore has $w_e(U) \leq -\varepsilon/2$.

For planes write $U = \ker M$, and put $A_\beta = \text{span}(L_\beta, X_2)$ and $B = \text{span}(X_1, X_2)$ in the dual space. For distinct conjugates β, β' ,

$$A_\beta \cap A_{\beta'} = A_\beta \cap B = \langle X_2 \rangle.$$

This follows by comparing the X_1 and X_3 coefficients. At a grouped infinite place the minimum pair has weight $(\rho - \varepsilon)/d$ unless $M \in A_\beta$; the finite minimum is $-1 - \rho$ unless $M \in B$. The following list is exhaustive. Proportional normals are identified.

Position of M	$w_e(\ker M)$
$M \notin B \cup \bigcup_\beta A_\beta$	$-1 - \varepsilon$
$M \in A_\beta \setminus (\langle L_\beta \rangle \cup \langle X_2 \rangle)$	$-1 - \varepsilon + (1 - \rho)/d$
$M \in \langle L_\beta \rangle$ for some β	$-(1 - 1/d)(1 + \varepsilon)$
$M \in B \setminus (\langle X_1 \rangle \cup \langle X_2 \rangle)$	$\rho - 1 - \varepsilon$
$M \in \langle X_1 \rangle \cup \langle X_2 \rangle$	$-\varepsilon$

Each entry is at most $-\varepsilon$, since $d \geq 2$, $\varepsilon \leq 1/2$ and $\rho \leq 1 - \varepsilon$. For example, at $M = L_\beta$ the pair for that conjugate must be (X_1, X_2) , whereas at $M = X_1$ the finite pair is (X_2, X_3) . These observations also cover the ties when $\rho = 0$. Equation (4.1) proves both bounds. $\square$

5 A finite-support moving-weight bound

We prove the arithmetic input separately. Throughout this section the number field K , the finite set S , the forms $\mathcal{L}$, and the finite place $v_0 \notin S$ are fixed. The dimension is three. We assume that

S contains every infinite place, that the forms outside S are the coordinate forms, and that their determinant at every place is 1. Write $s = |S|$. We call a system of real weights admissible if

$$c_{iv} = 0 \ (v \notin S), \quad \sum_i c_{iv} = 0, \quad \sum_v \max_i c_{iv} \leq 1, \quad w_c(U) \leq 0 \ (0 \neq U \subsetneq \overline{\mathbb{Q}}^3). \quad (5.1)$$

The last condition is the semistability established in Section 4 for the weights used in our application.

Theorem 5.1. *There are constants depending only on the fixed data with the following property. For $0 < \delta \leq 1$ there exist $\Omega(\delta) > 2$ and $C_*(\delta) > e$ with*

$$\Omega(\delta) \ll \delta^{-5}, \quad \log \log C_*(\delta) \ll \delta^{-2} \log(2/\delta), \quad (5.2)$$

such that any sequence $(c^{(h)}, Q_h)$ of admissible weights and parameters with

$$Q_1 \geq C_*(\delta), \quad Q_{h+1} > Q_h^{\Omega(\delta)}, \quad \{x \neq 0 : H_{\mathcal{L}, c^{(h)}, Q_h}(x) \leq Q_h^{-\delta}\} \neq \emptyset$$

has length $O(\delta^{-2})$.

Here is the structure of the proof. At each parameter a small twisted height produces a gap between successive infima. Semistability turns that gap into a height lower bound for an infimum subspace. An exterior power then gives two independent vectors with small local coordinates in a plane of large height. After selecting a common exterior power, we construct one auxiliary polynomial for all the selected parameters. A nonvanishing proposition supplies a derivative that is nonzero at grid points in the planes. Its local bounds give a product less than 1, contradicting the product formula.

We use the fixed-form height comparison and absolute Minkowski theorem from [3, Lemma 7.1 and Proposition 9.2], the determinant estimate from [3, Lemma 10.2], and the rescaling and Davenport construction from [3, Lemmas 11.1–11.3]. For the polynomial we use the number-field Siegel lemma [3, Lemma 13.1]; the nonvanishing input is [3, Proposition 12.1]. Below we give the determinant calculation, the simultaneous coefficient conditions with changing weights, the derivative estimates, and the cutoff calculation explicitly.

All constants are uniform in $c^{(h)}, Q_h, \delta$ once the fixed data have been chosen. The exterior forms will depend only on $\mathcal{L}$ and the exterior degree. The planes, vectors, and local exponents may depend on h . In particular the permutation arising from Davenport's construction changes the assignment of exponents, not the underlying forms. This distinction allows the coefficient equations to be imposed over K in fixed coordinate systems. We do not apply the final fixed-weight interval theorem of [3] separately at each point.

5.1 Pointwise data and the height of the exterior plane

We first give a pointwise version with a cutoff of order $\exp(O(1/\delta))$. This smaller cutoff will be used before imposing the height condition in the nonvanishing proposition.

Lemma 5.2 (Pointwise exterior data). *There are constants $A_0 > 0$ and $c_0 > 0$, depending only on $K, \mathcal{L}$, such that if c is admissible, $Q \geq \exp(A_0/\delta)$ and $H_{\mathcal{L}, c, Q}(x) \leq Q^{-\delta}$ for some nonzero x , then there is $k \in \{1, 2\}$ with the following properties. Put $N = \binom{3}{k} = 3$. There are fixed exterior forms $\hat{L}_{lv}$, depending only on $k, \mathcal{L}$, and independent vectors $z_1, z_2 \in \overline{\mathbb{Q}}^3$ spanning a plane T such that*

$$H_2(T) \geq Q^{c_0 \delta}. \quad (5.3)$$

For every place w of a common finite extension E/K ,

$$\|\widehat{L}_{lw}(z_j)\|_w \leq \begin{cases} Q^{d(w|v)a_{lv}}, & w \mid v \neq v_0, \\ Q^{d(w|v_0)b_l}, & w \mid v_0, \end{cases} \quad (5.4)$$

for $l = 1, 2, 3$ and $j = 1, 2$, where, with $\gamma_v = \max_i c_{iv}$,

$$\sum_l a_{lv} = 0, \quad |a_{lv}| \leq 2\gamma_v, \quad a_{lv} = 0 \ (v \notin S), \quad \sum_l b_l \leq -\delta/3, \quad |b_l| \leq 3. \quad (5.5)$$

Here H_2 is the absolute Euclidean height of the exterior coordinate vector defining a subspace, as in [3, Section 6].

Proof. The gap between successive infima. Let $\lambda_1 \leq \lambda_2 \leq \lambda_3$ be the successive infima of $H_{\mathcal{L},c,Q}$ over $\overline{\mathbb{Q}}$. Since $\sum_v \gamma_v \leq 1$, the fixed-form height comparison gives a constant $C_0 \geq 1$ such that

$$C_0^{-1}Q^{-1} \leq \lambda_1, \quad a_* := 3^{-3/2} \leq P_\lambda := \lambda_1\lambda_2\lambda_3 \leq 8. \quad (5.6)$$

The second assertion is the absolute Minkowski theorem; the determinant assumption makes its determinant factor equal to 1. The assumed small vector gives $\lambda_1 \leq Q^{-\delta}$. Hence $a_* \leq Q^{-\delta}\lambda_3^2$, so

$$\lambda_3 \geq a_*^{1/2}Q^{\delta/2} \geq 1 \quad \text{provided } Q^\delta \geq 3^{3/2}.$$

Consequently $(\lambda_1/\lambda_2)(\lambda_2/\lambda_3) \leq Q^{-\delta}$. For some $k \in \{1, 2\}$ we therefore have

$$g := \lambda_k/\lambda_{k+1} \leq Q^{-\delta/2} < 1. \quad (5.7)$$

By [3, Lemma 9.1], the points below any height strictly between λ_k and λ_{k+1} span the same k -dimensional subspace T_k , defined over K .

Put $A_k = \lambda_1 \cdots \lambda_k$. Monotonicity on the two sides of the gap gives

$$A_k^{3-k} \leq (g\lambda_{k+1})^{k(3-k)}, \quad (\lambda_{k+1} \cdots \lambda_3)^k \geq \lambda_{k+1}^{k(3-k)}.$$

Dividing these inequalities and using $k(3-k) = 2$ yields

$$A_k \leq P_\lambda^{k/3} g^{k(3-k)/3} \leq 8^{k/3} Q^{-\delta/3}. \quad (5.8)$$

The determinant calculation. Let $F_1, \dots, F_{R_0}$ be the distinct forms occurring in $\mathcal{L}$. They include the three coordinate forms, so $R_0 \geq 3$. Define the fixed quantity

$$\mathfrak{H} = \prod_{v \in M_K} \max_{1 \leq i_1 < i_2 < i_3 \leq R_0} \|\det(F_{i_1}, F_{i_2}, F_{i_3})\|_v \geq 1.$$

This is the form height in [3, Lemma 10.2]. Choose $0 < \eta < 1$ with $(1+\eta)\lambda_k < \lambda_{k+1}$ and independent $g_1, \dots, g_k \in T_k$ satisfying $H_{\mathcal{L},c,Q}(g_j) \leq (1+\eta)\lambda_j$. Such vectors can be chosen successively from the definition of the infima. Work in a finite extension containing their coordinates.

At each place v , select a restricted independent k -tuple $L_{i_1(v),v}, \dots, L_{i_k(v),v}$ that realizes the minimum in the definition of $w_c(T_k)$. For $w \mid v$ put

$$\theta_w = \det(L_{i_l(v),w}(g_j))_{l,j=1}^k, \quad H_{jw} = \max_i \|L_{iw}(g_j)\|_w Q^{-c_{iw}}.$$

These determinants are nonzero. Let $s(w)$ be the local degree divided by $[E : \mathbb{Q}]$ at infinite places and zero at finite places; then $\sum_w s(w) = 1$. The determinant expansion gives

$$\|\theta_w\|_w \leq (k!)^{s(w)} \prod_{j=1}^k H_{jw} Q^{\sum_{i=1}^k c_{i_l(v),w}}.$$

Since $\sum_{w|v} d(w | v) = 1$, multiplication gives

$$\begin{aligned} \prod_w \|\theta_w\|_w &\leq k! \prod_{j=1}^k H_{\mathcal{L},c,Q}(g_j) Q^{w_c(T_k)} \\ &\leq k! 2^k A_k \leq 32Q^{-\delta/3}. \end{aligned} \tag{5.9}$$

The middle inequality is where semistability is used.

There are at most $\binom{R_0}{k}$ distinct nonzero determinant values obtained from all restricted k -tuples of the F_i . Set $D_k = \binom{R_0}{k} - 1$ and $C_{\text{form}} = \sqrt{3}\mathfrak{H}$. The minimum-determinant estimate [3, Lemma 10.2, equation (10.2)] states that

$$\prod_w \min_{\theta \neq 0} \|\theta\|_w \geq (C_{\text{form}} H_2(T_k))^{-D_k}. \tag{5.10}$$

In particular the left side is a lower bound for the product in (5.9). Combining the two estimates gives

$$H_2(T_k) \geq C_{\text{form}}^{-1} 32^{-1/D_k} Q^{\delta/(3D_k)}.$$

With $D_* = \max(D_1, D_2)$, the condition $Q^{\delta/6} \geq 32C_{\text{form}}^{D_*}$ absorbs the fixed factor and gives

$$H_2(T_k) \geq Q^{\delta/(6D_k)} \geq Q^{c_0\delta}, \quad c_0 = \frac{1}{12R_0^3}. \tag{5.11}$$

Rescaling and exterior powers. Choose a full basis near the successive infima, with its first k vectors spanning T_k . The rescaling and Davenport construction in [3, Lemmas 11.1–11.3] gives, in a finite extension, a permutation π and a basis h_1, h_2, h_3 preserving these spans, with

$$\|L_{iw}(h_j)\|_w \leq 3^{-s(w)} Q^{c_{iw}} \quad (w \nmid v_0), \tag{5.12}$$

$$\|L_{\pi(i),w}(h_j)\|_w \leq (3^9 \min(\lambda_i, \lambda_j))^{d(w|v_0)} \quad (w | v_0). \tag{5.13}$$

For this use of Davenport's lemma only the local rescaling and the near-infimum basis are needed. Its proof chooses an arbitrarily small positive error satisfying the gap conditions in its equation (11.1). It does not use the large cutoff in the final interval theorem.

Put $p = 3 - k$, and list the p -element subsets I_1, I_2, I_3 of $\{1, 2, 3\}$ in lexicographic order. Define

$$\widehat{L}_{lv} = \bigwedge_{i \in I_l} L_{iv}, \quad a_{lv} = \sum_{i \in I_l} c_{iv}, \quad \widehat{h}_l = \bigwedge_{i \in I_l} h_i,$$

and retain $z_1 = \widehat{h}_1, z_2 = \widehat{h}_2$. For $k = 1$ these are $h_1 \wedge h_2, h_1 \wedge h_3$; for $k = 2$ they are h_1, h_2 . They are independent. Their span T satisfies $H_2(T) = H_2(T_k)$ by [3, Lemma 6.1], proving (5.3).

Away from v_0 , determinant expansion in (5.12) gives the factor $((p!)3^{-p})^{s(w)} \leq 1$, proving the ordinary bound in (5.4). Each original weight lies in $[-2\gamma_v, \gamma_v]$. When $p = 2$, its pair sum is the negative of the omitted weight. Thus $|a_{lv}| \leq 2\gamma_v$ in both cases; also $\sum_l a_{lv} = 0$ and $a_{lv} = 0$ outside S .

The exponents at the auxiliary place. Let $\nu_l = \prod_{i \in I_l} \lambda_i$, in the same order. From (5.13) and the exterior determinant expansion, the bounds for the retained vectors are, up to the permutation of rows,

$$(Q^{b_1}, Q^{b_2}, Q^{b_3}) = (3^{27} \nu_1, 3^{27} \nu_2, 3^{27} \nu_2).$$

The constant 3^{27} is a common upper bound for both exterior degrees. We check the product separately:

$$\begin{aligned} k = 1 : \quad (\nu_1, \nu_2, \nu_3) &= (\lambda_1 \lambda_2, \lambda_1 \lambda_3, \lambda_2 \lambda_3), \quad \nu_1 \nu_2^2 = P_\lambda^2 \lambda_1 / \lambda_2, \\ k = 2 : \quad (\nu_1, \nu_2, \nu_3) &= (\lambda_1, \lambda_2, \lambda_3), \quad \nu_1 \nu_2^2 = P_\lambda \lambda_2 / \lambda_3. \end{aligned}$$

Hence (5.6) and (5.7) give $Q^{\sum_l b_l} \leq 64 \cdot 3^{81} Q^{-\delta/2}$. If $Q^{\delta/6} \geq 64 \cdot 3^{81}$, then $\sum_l b_l \leq -\delta/3$.

For $p = 1, 2$ set $M_p = \max(C_0^p, 8C_0^{3-p}, 1)$. Every product of p distinct infima satisfies

$$M_p^{-1} Q^{-p} \leq \nu_l \leq M_p Q^{3-p}.$$

The lower bound uses the lower bound for each infimum in (5.6); the upper bound uses their total product and the lower bound for the complementary infima. If $Q \geq 3^{27} \max(M_1, M_2)$, it follows that $Q^{-3} \leq 3^{27} \nu_l \leq Q^3$. This proves $|b_l| \leq 3$.

For completeness the conditions used above are exactly

$$\begin{aligned} Q^\delta &\geq 3^{3/2}, & Q^{\delta/6} &\geq 32C_{\text{form}}^{D*}, \\ Q^{\delta/6} &\geq 64 \cdot 3^{81}, & Q &\geq 3^{27} \max(M_1, M_2). \end{aligned} \tag{5.14}$$

They hold for $Q \geq \exp(A_0/\delta)$, $0 < \delta \leq 1$, if we take A_0 larger than each of

$$\frac{3}{2} \log 3, \quad 6 \log(32C_{\text{form}}^{D*}), \quad 6 \log(64 \cdot 3^{81}), \quad \log(3^{27} \max(M_1, M_2)).$$

This completes the pointwise argument with constants depending only on the fixed forms and field. $\square$

5.2 One auxiliary polynomial for the changing weights

Suppose m selected parameters have the same k in Lemma 5.2, and label them $h = 1, \dots, m$. Their exterior forms are identical. Write a_{hlv}, b_{hl} for their exponents and put

$$\Gamma_v = \max_{1 \leq h \leq m} \gamma_{hv}, \quad \gamma_{hv} = \max_i c_{iv}^{(h)}. \tag{5.15}$$

Each γ_{hv} is nonnegative and at most 1, by admissibility. Consequently

$$\sum_{v \in S} \Gamma_v \leq s. \tag{5.16}$$

We use this bound rather than 1: the maximum over h is taken separately at each place, so it cannot be moved outside the sum. Ordinary vanishing conditions are needed only where $\Gamma_v > 0$.

For positive block degrees $r_1, \dots, r_m$, let $D_r = \sum_h r_h$ and

$$\mathcal{U}(r) = \{(j_{hl}) \in \mathbb{Z}_{\geq 0}^{mN} : \sum_l j_{hl} = r_h \text{ for each } h\}, \quad V = |\mathcal{U}(r)| = \prod_h \binom{r_h + N - 1}{N - 1}.$$

The elementary bound $\binom{r_h + N - 1}{N - 1} \leq (eN)^{r_h}$ gives

$$V \leq (eN)^{D_r}. \tag{5.17}$$

Proposition 5.3 (Simultaneous coefficient vanishing). *Let $0 < \alpha < 1$ and suppose*

$$m > 2\alpha^{-2} \log(2s + 2). \quad (5.18)$$

There is a nonzero polynomial P over K , homogeneous of degree r_h in the block $X_h \in K^N$, with the following properties. In its expansion $P = \sum_j d_j^{(v)} \prod_{h,l} \hat{L}_{lv}(X_h)^{j_{hl}}$, we have

$$d_j^{(v)} = 0 \quad \text{if } \sum_{h,l} j_{hl} a_{hlv} / r_h \geq 3m\alpha\Gamma_v, \quad v \in S, \Gamma_v > 0, \quad (5.19)$$

$$d_j^{(v_0)} = 0 \quad \text{if } \sum_{h,l} j_{hl} b_{hl} / r_h \geq -m\delta/9 + 3m\alpha, \quad (5.20)$$

$$H_2(P) \leq C_K A^{Dr}. \quad (5.21)$$

Here $C_K \geq 1$ and $A \geq 2$ are fixed, independent of m, r_h , the weights, and the parameters.

Proof. Counting the equations. Choose j uniformly from $\mathcal{U}(r)$. The blocks j_h are independent, and symmetry gives $\mathbb{E}(j_{hl}) = r_h/N$. For an ordinary place with $\Gamma_v > 0$, the variables

$$X_{hv} = r_h^{-1} \sum_l j_{hl} a_{hlv}$$

have mean zero and lie in $[-3\Gamma_v, 3\Gamma_v]$. Hoeffding's inequality [3, Lemma 13.2] therefore gives

$$\Pr \left(\sum_h X_{hv} \geq 3m\alpha\Gamma_v \right) \leq \exp \left(-\frac{2(3m\alpha\Gamma_v)^2}{m(6\Gamma_v)^2} \right) = e^{-m\alpha^2/2}.$$

At v_0 use $X_{h0} = r_h^{-1} \sum_l j_{hl} b_{hl}$. Its mean is $N^{-1} \sum_l b_{hl} \leq -\delta/9$, and its range is $[-3, 3]$. The event $\sum_h X_{h0} \geq -m\delta/9 + 3m\alpha$ thus has probability at most $e^{-m\alpha^2/2}$ as well.

Each coefficient required to vanish gives one homogeneous linear equation in the V coefficients of P in the original variables. Counting the equations separately in each coordinate system, their total number U satisfies

$$U \leq (s+1)V e^{-m\alpha^2/2} < V/2. \quad (5.22)$$

This counts equations, including any redundancies, rather than a union of forbidden monomial indices in different bases. Linear dependencies can only reduce the number of constraints.

Uniform heights of the rows. Write $Y_h = \mathcal{A}_v X_h$, where $\mathcal{A}_v$ is the matrix of the fixed exterior forms at v . The coefficient equations use the substitution $X_h = \mathcal{A}_v^{-1} Y_h$. For a place u of K put

$$C_u = \max(1, \|\text{entries of } \mathcal{A}_v^{-1}\|_u : v \in S \cup \{v_0\}), \quad C = \prod_u C_u.$$

The matrices are invertible and fixed, so C is finite and fixed. The special-place matrix is the identity. Here $s(u)$ is the local degree divided by $[K : \mathbb{Q}]$ at an infinite place of K , and is zero at a finite place, so $\sum_u s(u) = 1$. The absolute value of each entry of a coefficient row is at most $N^{Dr s(u)} C_u^{Dr}$, by the multinomial expansion. Since the row has V coordinates, its Euclidean height is at most

$$V^{1/2} (NC)^{Dr} \leq A_1^{Dr}, \quad A_1 = (eN)^{1/2} NC.$$

Thus changing weights changes which rows are selected, but does not change their height bound.

If $U = 0$, a coordinate monomial already suffices. Otherwise the number-field Siegel lemma [3, Lemma 13.1] supplies a nonzero coefficient vector with

$$\begin{aligned} H_2(P) &\leq C_K V^{1/2} \left(\prod_{u=1}^U H_2(\text{row}_u) \right)^{1/(V-U)} \\ &\leq C_K (eN)^{D_r/2} A_1^{D_r U/(V-U)} \leq C_K ((eN)^{1/2} A_1)^{D_r}. \end{aligned}$$

Taking $A \geq \max(2, (eN)^{1/2} A_1)$ proves (5.21). There are only two possible exterior degrees, so a common choice of these constants works for both. $\square$

5.3 Divided derivatives and their local coefficients

We differentiate in the original variables. For a multi-index i , put

$$P_i = \left(\prod_{h,l} i_{hl}! \right)^{-1} \partial^i P, \quad |i|_r = \sum_{h,l} i_{hl}/r_h.$$

The following lemma makes the loss in the vanishing conditions and the uniformity of the coefficient heights explicit.

Lemma 5.4 (Derivative estimates). *If $|i|_r \leq 2m\alpha$, the divided derivative has expansions*

$$P_i = \sum_j d_{i,j}^{(v)} \prod_{h,l} \widehat{L}_{lv}(X_h)^{j_{hl}}, \quad (5.23)$$

$$d_{i,j}^{(v)} = 0 \quad \text{if} \quad \sum_{h,l} j_{hl} a_{hlv}/r_h > 12m\alpha\Gamma_v, \quad v \in S, \quad (5.24)$$

$$d_{i,j}^{(v_0)} = 0 \quad \text{if} \quad \sum_{h,l} j_{hl} b_{hl}/r_h > -m\delta/9 + 12m\alpha, \quad (5.25)$$

$$\prod_v \max_j \|d_{i,j}^{(v)}\|_v \leq C_K A_2^{D_r}, \quad (5.26)$$

where $A_2 \geq 2$ is fixed. Every monomial in a nonzero derivative has block degree $r_h - \sum_l i_{hl}$.

Proof. In the coordinates $Y_h = \mathcal{A}_v X_h$, the chain rule reads

$$\frac{\partial}{\partial X_{ha}} = \sum_l (\mathcal{A}_v)_{la} \frac{\partial}{\partial Y_{hl}}.$$

It follows that a coefficient with index j in the derivative is a linear combination of original Y -coefficients with indices $j + q$, where $q_{hl} \geq 0$ and $\sum_l q_{hl} = \sum_l i_{hl}$ in each block. The ordinary weighted shift satisfies

$$\left| \sum_{h,l} q_{hl} a_{hlv}/r_h \right| \leq 2\Gamma_v |i|_r \leq 4m\alpha\Gamma_v \leq 6m\alpha\Gamma_v.$$

At the special place the same shift is at most $3|i|_r \leq 6m\alpha$ in absolute value. If either strict inequality in (5.24) or (5.25) holds, every contributing original coefficient therefore meets its vanishing condition in Proposition 5.3. This proves the assertions of vanishing. When $\Gamma_v = 0$, all ordinary weights at v are zero, so the strict inequality there cannot occur.

For the height estimate let a_P be the original coefficient vector of P . In the original variables, a divided derivative multiplies each coefficient by a product of binomial coefficients, bounded by 2^{D_r} at infinite places and by 1 at finite places. Passing to the Y -coordinates uses the fixed inverse matrix. At any place u and for any one of the relevant coordinate systems this gives

$$\max_j \|d_{i,j}^{(v)}\|_u \leq V^{s(u)}(2N)^{D_r s(u)} C_u^{D_r} \max_j \|(a_P)_j\|_u.$$

For systems outside $S \cup \{v_0\}$ the matrix is the identity, and the same estimate remains valid. Multiply this bound using the system assigned to each place. Since $\sum_u s(u) = 1$ and the sup height of a_P is at most $H_2(P)$, (5.17) gives

$$\prod_v \max_j \|d_{i,j}^{(v)}\|_v \leq (2eN^2C)^{D_r} H_2(P) \leq C_K (2eN^2CA)^{D_r}.$$

This proves (5.26) with a fixed A_2 . □

5.4 Degree selection and the nonvanishing condition

Proof of Theorem 5.1. Choose

$$\alpha = \frac{\delta}{1000(s+1)}, \quad m = \left\lfloor 2\alpha^{-2} \log(2s+2) \right\rfloor + 1, \quad \Omega = \frac{4m^2}{\alpha}. \quad (5.27)$$

We specify $C_*(\delta)$ below; in particular it will be at least $\exp(A_0/\delta)$. Lemma 5.2 therefore applies to every parameter. There are two possible values of k , so a sequence of at least $2m-1$ parameters has m with a common k . Separation is preserved on passing to this subsequence. Relabel its parameters $Q_1 < \dots < Q_m$ and its exterior data by h . All vectors can be placed in a common finite extension of K ; the normalized local bounds are unchanged by further extension.

Choose an arbitrarily large positive integer r_1 such that $r_1 \log Q_1 > \alpha^{-1} \log Q_m$ and $C_K \leq 2^{r_1}$. For $h = 1, \dots, m$ put

$$L_h = \frac{r_1 \log Q_1}{\log Q_h}, \quad r_h = \lceil L_h \rceil, \quad B = Q_1^{r_1}.$$

This agrees with the chosen integer r_1 at $h = 1$. Since $L_h > 1/\alpha$, we have $L_h \leq r_h < L_h + 1 < (1 + \alpha)L_h$. Also $L_h \leq r_1$, hence $r_h \leq r_1$. These facts give

$$B \leq Q_h^{r_h} < B^{1+\alpha}, \quad r_h \leq r_1, \quad \frac{r_h}{r_{h+1}} > \frac{\log Q_{h+1}}{(1+\alpha) \log Q_h} > \frac{\Omega}{1+\alpha} \geq \frac{2m^2}{\alpha}. \quad (5.28)$$

The last ratio is in the decreasing-degree direction required by [3, Proposition 12.1, equation (12.3)].

Construct P by Proposition 5.3. In each plane T_h take the grid

$$\mathcal{G}_h = \{u_1 z_{h1} + u_2 z_{h2} : u_j \in \mathbb{Z}, |u_j| \leq \lceil N/\alpha \rceil\}, \quad N = 3.$$

It contains a grid of the size N/α required by that nonvanishing proposition. Set

$$E_m = (N-1)(3m^2/\alpha)^m, \quad A_3 = 2eA.$$

By (5.21) and $C_K \leq 2^{r_1} \leq 2^{D_r}$, we have $e^{D_r} H_2(P) \leq A_3^{D_r}$. The remaining hypothesis of the proposition is

$$H_2(T_h)^{r_h} \geq (e^{D_r} H_2(P))^{E_m} \quad (h = 1, \dots, m). \quad (5.29)$$

Its two sides have the bounds

$$H_2(T_h)^{r_h} \geq Q_h^{c_0 \delta r_h} \geq B^{c_0 \delta}, \quad (e^{D_r} H_2(P))^{E_m} \leq A_3^{D_r E_m} \leq A_3^{mr_1 E_m}.$$

It is therefore sufficient that

$$c_0 \delta \log Q_1 \geq m E_m \log A_3. \quad (5.30)$$

Under this condition [3, Proposition 12.1] supplies $x_h \in \mathcal{G}_h \setminus \{0\}$ and a multi-index with $|i|_r \leq 2m\alpha$ for which

$$P_i(x_1, \dots, x_m) \neq 0.$$

We now estimate this nonzero algebraic number at every place.

The grid factor. At an infinite place w , the triangle inequality introduces at most the sum of the absolute values of the two grid coefficients. In general, with $N - 1$ coefficients, this sum is at most

$$(N - 1) \lceil N/\alpha \rceil \leq (N - 1)(N/\alpha + 1) \leq N^2/\alpha,$$

because $\alpha \leq 1$. At finite places the coefficients are integers and the ultrametric inequality introduces no factor. Thus (5.4) gives

$$\|\widehat{L}_{lw}(x_h)\|_w \leq (N^2/\alpha)^{s(w)} Q_h^{d(w|v)a_{hlv}} \quad (w \mid v \neq v_0). \quad (5.31)$$

At $w \mid v_0$ use b_{hl} instead, with no grid factor since v_0 is finite. The factor N^2/α will be included in the final bound.

Ordinary-place exponents. Put $\kappa_h = r_h \log Q_h / \log B$, so $1 \leq \kappa_h < 1 + \alpha$. For a surviving monomial in the derivative at $w \mid v \neq v_0$, set

$$A_{hw} = \frac{1}{r_h} \sum_l j_{hl} d(w \mid v) a_{hlv}, \quad \Gamma_w = d(w \mid v) \Gamma_v.$$

The block degrees are at most r_h , so $|A_{hw}| \leq 3\Gamma_w$. The parameter contribution to this monomial is exactly

$$\prod_h Q_h^{r_h A_{hw}} = B^{\sum_h \kappa_h A_{hw}}.$$

Using the derivative vanishing condition and then the ceiling error,

$$\sum_h A_{hw} \leq 12m\alpha\Gamma_w, \quad \left| \sum_h (\kappa_h - 1) A_{hw} \right| \leq 3m\alpha\Gamma_w.$$

The ordinary-place exponent is therefore at most $15m\alpha\Gamma_w$. This also holds outside S , where all the ordinary exponents vanish.

The exponent at v_0 . Here define $A_{hw} = d(w \mid v_0) \sum_l j_{hl} b_{hl} / r_h$. We have $|A_{hw}| \leq 3d(w \mid v_0)$, while (5.25) gives

$$\sum_h A_{hw} \leq d(w \mid v_0)(-m\delta/9 + 12m\alpha).$$

The ceiling error is at most $3m\alpha d(w \mid v_0)$. The special-place exponent is thus at most $d(w \mid v_0)(-m\delta/9 + 15m\alpha)$.

Multiplication over all places. A derivative has at most $V \leq (eN)^{D_r}$ monomials. Locally the sum introduces the factor $V^{s(w)}$, and the grid contributes at most $(N^2/\alpha)^{D_r s(w)}$. Multiply these with (5.26). The identities

$$\sum_w s(w) = 1, \quad \sum_{w|v} d(w \mid v) = 1, \quad \sum_{v \in S} \Gamma_v \leq s$$

give

$$\begin{aligned} \prod_w \|P_i(x_1, \dots, x_m)\|_w &\leq C_K (A_2 e N^3 / \alpha)^{D_r} B^m [-\delta/9 + 15(s+1)\alpha] \\ &\leq G(\alpha)^{D_r} (B^m)^{-\delta/12}, \quad G(\alpha) = 2A_2 e N^3 / \alpha. \end{aligned} \quad (5.32)$$

Here $C_K \leq 2^{D_r}$, and our choice of α ensures

$$-\delta/9 + 15(s+1)\alpha = -\delta/9 + 15\delta/1000 \leq -\delta/12.$$

Since $D_r \leq mr_1$ and $B = Q_1^{r_1}$, the logarithm of the right side of (5.32) is at most

$$mr_1 (\log G(\alpha) - (\delta/12) \log Q_1).$$

It is strictly negative if

$$\log Q_1 \geq 24\delta^{-1} \log G(\alpha). \quad (5.33)$$

This contradicts the product formula for the nonzero value supplied by the nonvanishing proposition.

The uniform cutoff. To meet the pointwise cutoff, (5.30), and (5.33) simultaneously, take

$$C_*(\delta) = \exp \left(\frac{A_0 + 1}{\delta} + \frac{2mE_m \log A_3}{c_0 \delta} + \frac{24 \log G(\alpha)}{\delta} \right). \quad (5.34)$$

The dependence on δ can now be checked without suppressing the nonvanishing exponent. With the fixed support, we have $\alpha \asymp \delta$, $m \asymp \delta^{-2}$, and consequently $\Omega = 4m^2/\alpha \ll \delta^{-5}$. Moreover

$$\log E_m = \log(N-1) + m \log(3m^2/\alpha) \ll \delta^{-2} \log(2/\delta). \quad (5.35)$$

Taking the logarithm of the sum defining $\log C_*$, and bounding it by the largest summand up to a fixed factor, gives

$$\log \log C_* \ll 1 + \log m + \log E_m + \log(1/\delta) + \log(1 + \log G(\alpha)) \ll \delta^{-2} \log(2/\delta).$$

All fixed factors are absorbed into constants depending only on the fixed data. A sequence of length $2m-1$ has been ruled out, so its length is at most $2m-2 = O(\delta^{-2})$. This proves the theorem. $\square$

Corollary 5.5. *Under the same hypotheses, suppose $Q_{h+1} > Q_h^2$. The number of parameters at or above $C_*(\delta)$ is $O(\delta^{-2} \log(2/\delta))$.*

Proof. Discard any parameters below $C_*(\delta)$ and set $q = \lceil \log_2 \Omega \rceil \ll \log(2/\delta)$. Partition the remaining indices into their q residue classes. Successive parameters in any one class are q steps apart, and iteration of the strict doubling inequality gives $Q_{h+q} > Q_h^{2^q} \geq Q_h^\Omega$. Theorem 5.1 bounds each class by $O(\delta^{-2})$; summing the q bounds proves the assertion. $\square$

The logarithm in this corollary comes from the separation partition. The logarithm in (5.2) comes from (5.35). These are distinct conditions, and both will be retained in the dyadic counting argument below.

6 Dyadic scales and fixed-quality layers

Fix the forms and support from Section 3, and choose a finite place v_0 above a rational prime not dividing b . The admissibility requirements in Section 5 follow from (3.8) and Proposition 4.1.

Lemma 6.1 (Counting all prefix scales). *There are constants $K_0, K_1 > 0$, depending only on ξ, b , with the following property. Let $B \geq 2$ and let I be a finite set of integers. For each $i \in I$, suppose there is an approximation satisfying (2.1)–(2.2), with $H = 2^i$, such that*

$$t_i \geq T, \quad v_i \geq 2^i/B.$$

If

$$\log T \geq K_0 B^2 \log(2B), \tag{6.1}$$

then

$$|I| \leq K_1 B^2 \log(2B). \tag{6.2}$$

Proof. Put $\tau_i = \lfloor \log_2 t_i \rfloor$ and $\mu_\tau = \#\{i \in I : \tau_i = \tau\}$. For each nonempty class retain its largest index i_τ and that index's approximation. Since $t_i \leq 2^i$, every member satisfies $i \geq \tau$. The indices are distinct, so $i_\tau \geq \tau + \mu_\tau - 1$. As $t_{i_\tau} < 2^{\tau+1}$, this yields

$$\frac{v_{i_\tau}}{t_{i_\tau}} > \frac{2^{i_\tau - \tau - 1}}{B} \geq \frac{2^{\mu_\tau - 2}}{B}. \tag{6.3}$$

For each integer $j \geq 1$ set

$$\mathcal{T}_j = \{\tau : \mu_\tau \geq j\}, \quad N_j = |\mathcal{T}_j|, \quad \varepsilon_j = \frac{2^{j-3}}{B}, \quad \delta_j = \frac{\varepsilon_j}{3 + \varepsilon_j}.$$

Only levels with $N_j > 0$ need consideration. At such a level, (6.3) and $v_i/t_i \leq 1$ imply $2^{j-2}/B < 1$, hence $0 < \varepsilon_j < 1/2$. Every retained approximation in that level has $v_i/t_i > 2\varepsilon_j$, so (3.9) applies with the common ε_j and δ_j .

Partition $\mathcal{T}_j$ according to the parity of τ . In either part consecutive values differ by at least two. Thus their retained parameters satisfy $t_{\text{next}} \geq 2^{\tau+2} > 2t_{\text{previous}}$, and consequently $Q_{\text{next}} > Q_{\text{previous}}^2$.

We verify the cutoff uniformly over all nonempty levels. We have $\delta_j \geq \delta_1 = 1/(12B + 1)$. The function $\delta^{-2} \log(2/\delta)$ decreases on $(0, 1]$; therefore (5.2) gives

$$\log \log C_*(\delta_j) \ll B^2 \log(2B).$$

On the other hand, $Q = b^{D_{\varepsilon_j} t}$ and $D_{\varepsilon_j} \geq 1$ give $\log \log Q \geq \log T + \log \log b$. Increasing K_0 in (6.1) puts every retained point above its level's cutoff. Monotonicity of C_* itself is not needed.

Apply Corollary 5.5 to both parity classes. Since $\delta_j \geq 2\varepsilon_j/7$, it follows that

$$N_j \ll \delta_j^{-2} \log(2/\delta_j) \ll B^2 4^{-j} \log(2B). \tag{6.4}$$

The constants are uniform in B, j and the approximations. Summation over multiplicity levels now gives

$$|I| = \sum_{\tau} \mu_\tau = \sum_{j \geq 1} N_j \ll B^2 \log(2B) \sum_{j \geq 1} 4^{-j} \ll B^2 \log(2B),$$

as required. $\square$

Remark 6.2. Each application of Corollary 5.5 uses a fixed quality δ_j . Changing the layer changes this common quality and the associated normalized heights. The uniform bound for the cutoffs makes this legitimate; no theorem for a chain with varying smallness exponents δ_h is used.

7 Window bounds and the endpoint

Proof of Theorem 1.1. Fix $\theta > 1$, let $Y = X^\theta$, and put $F(x) = \sqrt{\log x / \log \log x}$. Suppose that, throughout the integer lengths in $[X, Y]$,

$$R(k) \leq CkF(k), \quad (7.1)$$

where $C > 0$ is a fixed constant to be chosen sufficiently small. Let $F_Y = F(Y)$, $B = 24CF_Y$, and

$$I = \{i \in \mathbb{Z} : 8CF_Y X \leq 2^i \leq Y\}.$$

For sufficiently large X , $B \geq 2$ and $CF_Y \geq 1$. Lemma 2.2 provides the approximations needed in Lemma 6.1, with $T = X/3$. The window buffer ensures that every invoked value of $R(k)$ lies inside $[X, Y]$.

For fixed C, θ , the number of scales satisfies

$$|I| = \frac{\log(Y/X)}{\log 2} - O(\log F_Y + |\log C| + 1) = \left(\frac{\theta - 1}{\log 2} + o(1) \right) \log X. \quad (7.2)$$

Also

$$B^2 \log(2B) = (288C^2 + o(1)) \log Y = (288\theta C^2 + o(1)) \log X. \quad (7.3)$$

Choose $C > 0$ so small that

$$288\theta K_0 C^2 < \frac{1}{2}, \quad 288\theta K_1 C^2 < \frac{\theta - 1}{2 \log 2}.$$

Then $\log(X/3) \geq K_0 B^2 \log(2B)$ for sufficiently large X , and Lemma 6.1 applies. Its bound (6.2) contradicts (7.2)–(7.3). Hence some integer $L \in [X, X^\theta]$ has $R(L) > CLF(L)$. By (1.1), the same L has $p(L) \geq R(L) - L > (C/2)LF(L)$ once X is large enough. This proves the theorem with $c = C$. $\square$

Proof of Corollary 1.2. Take, for example, $\theta = 3/2$ in Theorem 1.1. The lengths provided tend to infinity with X , proving the positive limsup. Multiplying the resulting lower bounds by $(\log \log L)^{\gamma-1/2}$ proves the first assertion in (1.3); multiplying by $(\log L)^{1/2-\eta}/\sqrt{\log \log L}$ proves the second. $\square$

Proof of Theorem 1.3. For the given F , we have $F(X) \rightarrow \infty$ and

$$F(X)^2 \log(2F(X)) = o(\log X). \quad (7.4)$$

Fix $C_0 > 0$ and write

$$W = AF(X)^2 \log(2F(X)), \quad Y = Xe^W.$$

For fixed A , equation (7.4) gives $W = o(\log X)$. Thus $\log Y / \log X \rightarrow 1$. Substitution into the definition of F then yields $F(Y)/F(X) \rightarrow 1$.

Suppose $R(k) \leq C_0 k F(k)$ at all integer lengths in $[X, Y]$. Apply Lemma 2.2 with $B = 24C_0 F(Y)$ and $I = \{i : 8C_0 F(Y) X \leq 2^i \leq Y\}$. Since $W / \log F(X) \rightarrow \infty$, we have

$$|I| = \left(\frac{A}{\log 2} + o(1) \right) F(X)^2 \log(2F(X)), \quad B^2 \log(2B) = (576C_0^2 + o(1)) F(X)^2 \log(2F(X)).$$

The threshold (6.1), with $T = X/3$, holds by (7.4). Choose $A > 576K_1 C_0^2 \log 2$. For all sufficiently large X , (6.2) is then a contradiction. Thus some $L \in [X, Y]$ satisfies $R(L) > C_0 L F(L)$. To obtain the assertion for the prescribed C , apply this with $C_0 = 2C$. For large X , (1.1) gives $p(L) > CLF(L)$ at the same L , as well as $R(L) > CLF(L)$. Finally (7.4) gives $Y = X^{1+o(1)}$. $\square$

8 Scope and the remaining losses

The arithmetic proof uses finite common support to impose simultaneous coefficient vanishing with $m = O(\delta^{-2})$, and an explicit check of semistability for the full family of exact weights. In particular, the moving-weight theorem is proved in Section 5; it does not follow by simply applying the published fixed-weight theorem at each point. Nonreal conjugates and composite bases are handled by the mass and prime-weight identities in Section 3.

The dyadic argument records multiplicity across prefix scales. It does not give an arithmetic bound for all pairs of equal words within one prefix. If f_w counts the occurrences of a length- k word in a prefix of length H , then Cauchy–Schwarz gives

$$\sum_w f_w^2 \geq \frac{(H - k + 1)^2}{p(k)}.$$

The repetition lemma only needs one pair; multiplicities of arbitrary pairs are not retained by the subsequent arithmetic construction. Lemma 6.1 instead retains the multiplicity of prefix scales through the quality gain (6.3).

The exponent $1/2$ in the present estimates comes from (5.18), together with $\alpha \asymp \delta$ in (5.27). Two additional restrictions each contribute one logarithm at the final scale: the nonvanishing threshold has $\log \log C_* \ll \delta^{-2} \log(2/\delta)$, and passing from Ω -separation to ordinary doubling separation costs $q \asymp \log(2/\delta)$ in Corollary 5.5. These are separate requirements at the same order $B^2 \log(2B)$; their logarithmic costs are not multiplied in the dyadic proof. Both requirements must be addressed to remove the remaining $\sqrt{\log \log L}$ denominator by this method. This identifies limitations of the argument given here, without asserting an obstruction to all other methods.

The endpoint proof gives fixed-power windows with constants depending on their exponent. The proof of shrinking relative logarithmic windows uses (7.4); it does not establish such windows at $u = \gamma = 1/2$. No conclusion of positive entropy, disjunctivity, or normality follows from these block-complexity lower bounds.

References

- [1] B. Adamczewski and Y. Bugeaud, On the complexity of algebraic numbers I. Expansions in integer bases, *Ann. of Math. (2)* **165** (2007), no. 2, 547–565. [doi:10.4007/annals.2007.165.547](https://doi.org/10.4007/annals.2007.165.547).
- [2] Y. Bugeaud and J.-H. Evertse, On two notions of complexity of algebraic numbers, *Acta Arith.* **133** (2008), no. 3, 221–250. [doi:10.4064/aa133-3-3](https://doi.org/10.4064/aa133-3-3).
- [3] J.-H. Evertse and R. G. Ferretti, A further improvement of the quantitative Subspace Theorem, *Ann. of Math. (2)* **177** (2013), no. 2, 513–590. [doi:10.4007/annals.2013.177.2.4](https://doi.org/10.4007/annals.2013.177.2.4).
- [4] Y. Bugeaud and D. H. Kim, A new complexity function, repetitions in Sturmian words, and irrationality exponents of Sturmian numbers, *Trans. Amer. Math. Soc.* **371** (2019), no. 5, 3281–3308. [doi:10.1090/tran/7378](https://doi.org/10.1090/tran/7378).